

Rolle's & MVT Theorem

A. Verify the hypothesis of Rolle's Theorem for the following functions:

1. $f(x) = x^2 - 6x + 8$; $[2, 4]$

2. $f(x) = \cos x$; $[\frac{\pi}{2}, \frac{3\pi}{2}]$

3. $f(x) = \frac{1}{2}x - \sqrt{x}$; $[0, 4]$

B. Verify the hypothesis of Mean value theorem for the following functions:

1. $f(x) = x^3 + x - 4$; $[-1, 2]$

2. $f(x) = \sqrt{x+1}$; $[0, 3]$

3. $f(x) = \sqrt{25 - x^2}$; $[0, 5]$